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Research Article

UNKNOWN AGENT IN MARES: *CHLAMYDIA ABORTUS*

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ABSTRACT. *Chlamydia abortus* is a gram-negative, obligate intracellular pathogen. The causative agent causes infections by affecting the urogenital system in humans and many animals due to its zoonotic character. It has been reported that *C. abortus* previously developed forms characterized by abortion in sheep and goats. Studies on horses from farm animals are very rare in our country and in the world. The aim of this study was to investigate the molecular prevalence of *C. abortus* in mares. For our research, vaginal swab samples were collected from 123 mares from farms in different settlements in Turkey. Samples were tested by real time PCR method. As a result of the test, a total of 4 (3.25%) samples were evaluated as *C. abortus* positive. As a result of the evaluation, in the analysis of variance with the 95% confidence interval, it was determined as 1.0577 for Marmara, 1.0233 for Southeast Anatolia, and 1.0000 for the Eastern Anatolia region. The presence of this agent, which is very difficult in culture isolation, was demonstrated in mares by molecular diagnosis. According to this result, it was concluded that it would be beneficial to take precautions against *C. abortus* infections in horse farms as in other farm animals. At the same time, our study is one of the rare studies in which *C. abortus* infection was investigated by real time PCR in mares genital swap samples. It will also contribute to further studies to determine the presence of infection in mares to understand the transmission cycle. It should not be forgotten that this agent, which is zoonotic and causes abortions, may also carry a risk for people working in horse farms and having a history of contact.

Keywords: *Abortion, Chlamidya abortus, mare, real time PCR, zoonoses.*

INTRODUCTION

Chlamydophila abortus causes respiratory and reproductive infections in humans and animals. It is also a zoonotic agent. Etiologically, it is a Gram-negative, obligate intracellular pathogen [1, 2].

This infection, which is seen all over the world, causes abortion in cattle, sheep, goats, pigs and horses. The agent particularly affects the placenta [3]. Due to its zoonotic importance, it threatens human health also. Especially in pregnant women who come into contact with infected meat, the rate of infection is high. Also, farm workers are known to be at high risk. Animals infected with *C. abortus* shed the agent in faeces, milk, vaginal and placental waste [3]. This infection is characterized by atypical pneumonia, catarrhal manifestations in humans. There are also reports of reproductive problem *Chlamydia trachomatis* associated with pregnant women [4]. In addition, due to the deadly consequences of human psittacosis caused by *Chlamydia psittaci*, laws are enacted in some countries that require reporting [5, 6]. *C. psittaci* is the most common cause of bird origin. However, recently, detection of *C. abortus*, previously classified as serotype 1 of *Chlamydia psittaci*, in ducks and turkeys has attracted attention [6]. Therefore, it has been included in the classification as a separate species of Chlamydiaceae. According to these laws and regulations, the importance of research on the existence of chlamydial species other than *C. psittaci* is emphasized. Due to the genetic similarity between these two species, it is important to make an accurate and decisive definition of abortion [6]. Especially since both *C. abortus* and *C. psittaci* are zoonotic agents, it is very important to understand the risk of exposure of horses to abortion tissues. Usually, the diagnosis of the agent can be made by taking samples of post-abortion vaginal discharge. Therefore, serious protection and biosecurity measures should be taken after abortion [5]. Recent studies have shown that in horses c. emphasizes the need for detecting the presence of abortion [1, 2]. In our study, we investigated the presence of *C. abortus* in mare vaginal swab samples with real time PCR, one of the molecular methods.

MATERIALS AND METHODS

Sampling and Study Material

Vaginal swab samples of mares with abortion collected from different region of Turkey. The samples represented the three geographic regions of Turkey. Distribution of the number of samples by regions; Marmara region was recorded as 52, Southeast Anatolia region 43 and Eastern Anatolia region 28 [Figure 1].

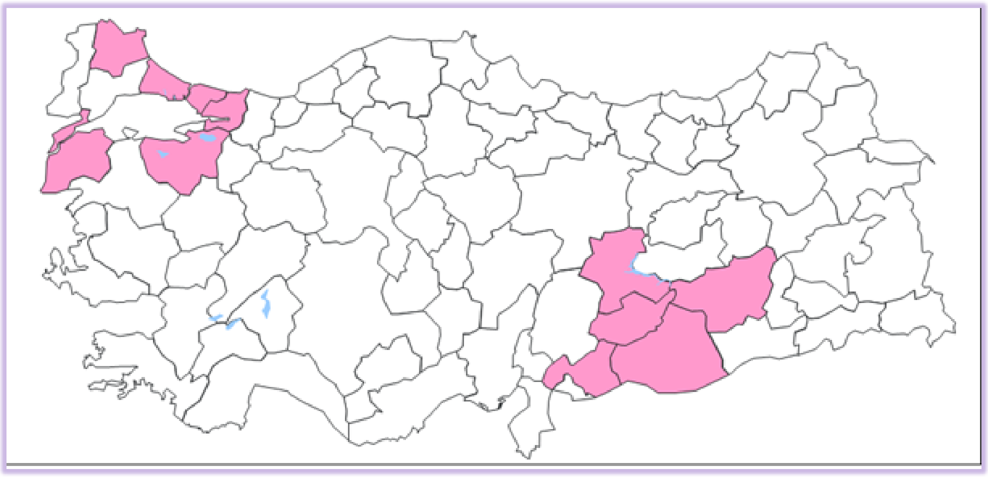

Fig. 1. *Geographic regions where samples were collected*

The mares randomly selected from the farms were older than 36 months. In this context, 123 mare vaginal swabs samples were analysed. Vaginal swabs were first washed with an antiseptic solution, and samples were taken from mares by providing hygienic conditions. Vaginal swabs contained transport medium. There were no major clinical signs at the time of specimen collection. After the samples were collected, they were immediately sent to the laboratory. Samples were stored at -20 °C until analysis.

DNA extraction

DNA extraction was performed using a commercial extraction kit [Qiagen, Germany] according to the instructions reported by the manufacturer.

Real time PCR analysis

Real time PCR assay was performed from all DNA samples targeting the ompA gene, as previously reported by Panchev *et al.* [6]. Mare vaginal swab samples were screened to detect the presence of *C. abortus* DNA [Table 1]. Real-time PCR analyses were performed on the LightCycler96 [Roche, Germany] instrument. All analyses were performed using the positive and negative control (Necmettin Erbakan university faculty of veterinary medicine, microbiology department culture collection). Real-Time PCR analysis results were evaluated according to the amplification curves and Ct [threshold value cycle] data, together with positive and negative controls. Amplification observed in a sigmoidal curve and Ct ≤ 37 was considered direct positive. During the analysis of all samples, it was repeated twice tested. As a result of our testing, any amplification no curve observed or those with a Ct value [threshold cycle] of more than 37 were evaluated as a negative result in terms of *C. abortus*. The primers and probes used and the cycle conditions were applied as given in Table 1.

Table 1. *Real time PCR Cycling conditions, target gene and primer sequences*

Chlamydia abortus [Pantchev et al. 2009**]**	
Forward [CpaOMP1-F]	GCA-ACT-GAC-ACT-AAG-TCG-GCT-ACA- 3’
Reverse [CpaOMP1-R]	ACA-AGC-ATG-TTC-AAT-CGA-TAA-GAG-A- 3’
Taqman Probe [CpaOMP1-S]	TAA-ATA-CCA-CGA-ATG-GCA-AGT-TGG-TTT-AGC-G
Cycling Conditions	95°C/10 min 45 × [95°C/15 sec, 60°C/60 sec 45x [95-10sn, 57-30sn, 72-1sn

Statistical analysis

Statistical analysis of data IBM SPSS Statistic 22.0 carried out with the program.

RESULTS AND DISCUSSION

As a result of our study, 123 mare vaginal swab samples were taken from mares from different farms. Samples were tested by real time PCR method. As a result of the test, a total of 4 [3.25%] samples were evaluated as positive for *C. abortus* [Figure 1]. As a result of the evaluation, in the analysis of variance (1 value was given as negative, 2 values were positive) at the 95% confidence interval, it was determined as 10577 for Marmara, 10233 for southeast Anatolia, and 10000 for the Eastern Anatolia region. It is seen that the rate of chlamydia seen in mares in the Marmara region is higher than in other regions. However, although it seems to be statistically insignificant, it was concluded that the number of statistical significances would change when the number of samples was increased.

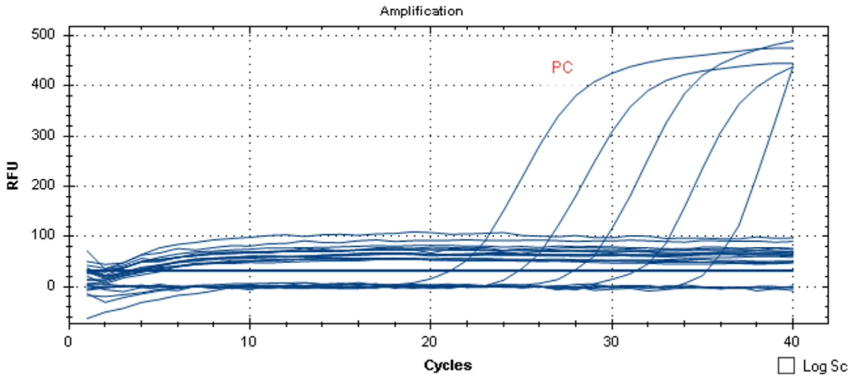

Fig. 2. *Amplification curves of C. abortus positive samples by real time PCR [pc: C. abortus positive control]*

For chlamydia, cell culture methods are used. However, difficulties such as low sensitivity, difficulties in standardization, and laborious construction process in laboratory conditions are reported. Apart from cell culture, there are commercial kits that detect chlamydia-specific antigen. The sensitivity of these tests is controversial. They are generally used as a screening test. [7]. For these reasons, recently, PCR tests, which can reach sensitive, reliable, high and fast results, are currently used for the diagnosis of these and similar special agents [8]. With this method, it was thought that early diagnosis would be provided for *Chylamidya abortus* in our study before the development of an occult infection or clinical picture. Reducing the risk of infection will also ensure that precautions are taken against the disease [3, 8]. With real-time PCR, contaminations that may develop in the laboratory environment can be prevented [4].

A high percentage of positivity was found in animal samples, especially in molecular studies [9, 10]. This organism reproductive losses in other animal species are also suspected. These animals, pigs and horses [11]. There may be more than one cause of embryonic deaths and urogenital system infections. In order to understand this situation adequately, each animal species and its effects on humans should be investigated [11]. Transmission between animals can develop in different ways. Contamination of pastures and pastures of animals, contact with contaminated materials are among the natural transmission routes of *C. abortus*. Transmission from animals to humans can occur through uterine discharge, milk, urine, and faeces [11,12]. Generally, there are studies on *C. abortus* agent related to sheep and goats. These studies are determined both by serology, cell culture and molecular methods [12]. Chlamydial abortion poses a great risk, especially in pregnant women [13]. In particular, contact of pregnant women with farm animals was determined when placental infection and foetal deaths were investigated [12, 13]. There are very limited studies in our country and in the world on the diagnosis of *C. abortus* by molecular studies from the urogenital sample of the mares we mentioned in our study. Baumann et al. [14] In Switzerland, 1.2% positive Chlamydiaceae positivity was detected by real time PCR in 162 fetal membrane samples. In parallel with our study, the prevalence of chlamydial species in equine abortion cases was found to be low in this study. It would be appropriate to consider that these *C. abortus* infections seen in horses may develop by exposure to infected ruminants such as sheep, which are more susceptible to the disease. At the same time, a different possible scenario is the cycle of contamination with possible contact of horse farms and wildlife [15, 16].

Ricard et al. In their study in Western Canada, 22 of 95 cases were determined as *C. abortus* by conventional PCR test Ricard et al [17]. detected *C. abortus* DNA in 6 of 50 mare samples. However, only 1 of these mares had abortion cases [18]. Although abortion cases did not develop in mares, cytological examinations performed before and after treatment in mares with positive *C. abortus* suggested that *C. abortus* may cause chronic endometritis in these animals [18]. In our study, it was reported by the breeders that they encountered urogenital system infections from time to time while taking mare vaginal swab samples from different farms and that the animals were treated with antibiotics. But no abortion cases were reported. Although no abortion cases were reported, positivity was found in 4 of 123 mare samples, suggesting that *C. abortus* may also cause different urogenital system problems [18]. The relationship between *C. abortus* and chronic endometritis is waiting to be clarified in future studies.

CONCLUSION

Considering the regions, there are significant seasonal differences between Southeastern and Eastern Anatolia in the Marma region. There are studies related to the survival rate and transmission of chlamydial agents in animals in different climatic conditions [19]. It has been reported that the survival of Chlamydia will increase and more positivity will be obtained in regions with warm temperatures and suitable precipitation [19]. In our study, 3 positives were found in the Marmara region, 1 positive in the Southeast Anatolian region, and no positivity was found in the samples in the Eastern Anatolia region. Although this situation gives a parallel view with other studies, it should not be considered significant since the highest number in the random sample was from the Marmara region [n=52]. When the results of the obtained studies are evaluated, it is recommended to evaluate seasonal differences in further studies with more data.

All these studies and the result of our research showed that *C. abortus* agent, which have a possible role in genital tract infections of mares, should not be ignored. When evaluating *Chlamydia abortus* infection, other bacterial and viral agents and non-infectious factors should be fully evaluated. It is recommended to carry out studies especially on contact with horse breeders. It is thought that research using this and similar advanced diagnostic methods will clearly reveal the *C. abortus* infection in horses.

Conflict of Interest. The authors declared that there is no conflict of interest.

Authorship Contributions. Derya Karataş Yeni: Investigation, Methodology, Supervision, Validation, Formal analysis, Visualization. Aslı Balevi: Visualization, Writing – review & editing. All authors read, revised, and approved the submitted manuscript.

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